```
public class Caramel extends SilkOreo{
  String texture = "Softy";
  public void method1() {
    System.out.println("Caramel m1");
  }
  public void method4() {
    System.out.println("Caramel m4");
  }
  public String toString(){
    method2();
    return "Caramel is "+ texture;
  }
}
public class Chocolate{
  String texture = "Chocolaty";
  public void method1() {
    method2();
    System.out.println("Chocolate m1");
  }
  public void method2() {
    System.out.println("Chocolate m2");
  }
  public String toString(){
    method2();
    return "Chocolate is "+ texture;
  }
}
public class DairyMilk extends Chocolate{
  String texture = "Yummy";
  public void method2() {
    System.out.println(this.texture);
    System.out.println("DairyMilk m2");
  }
  public void method3() {
    System.out.println("DairyMilk m3");
  }
}
public class KitKat extends Chocolate{
  String texture = "Crunchy";
  public void method1() {
    System.out.println("KitKat m1");
  }
  public void method4() {
    System.out.println("KitKat m4");
  }
  public String toString(){
    method2();
    return "KitKat is "+ texture;
  }
}
```

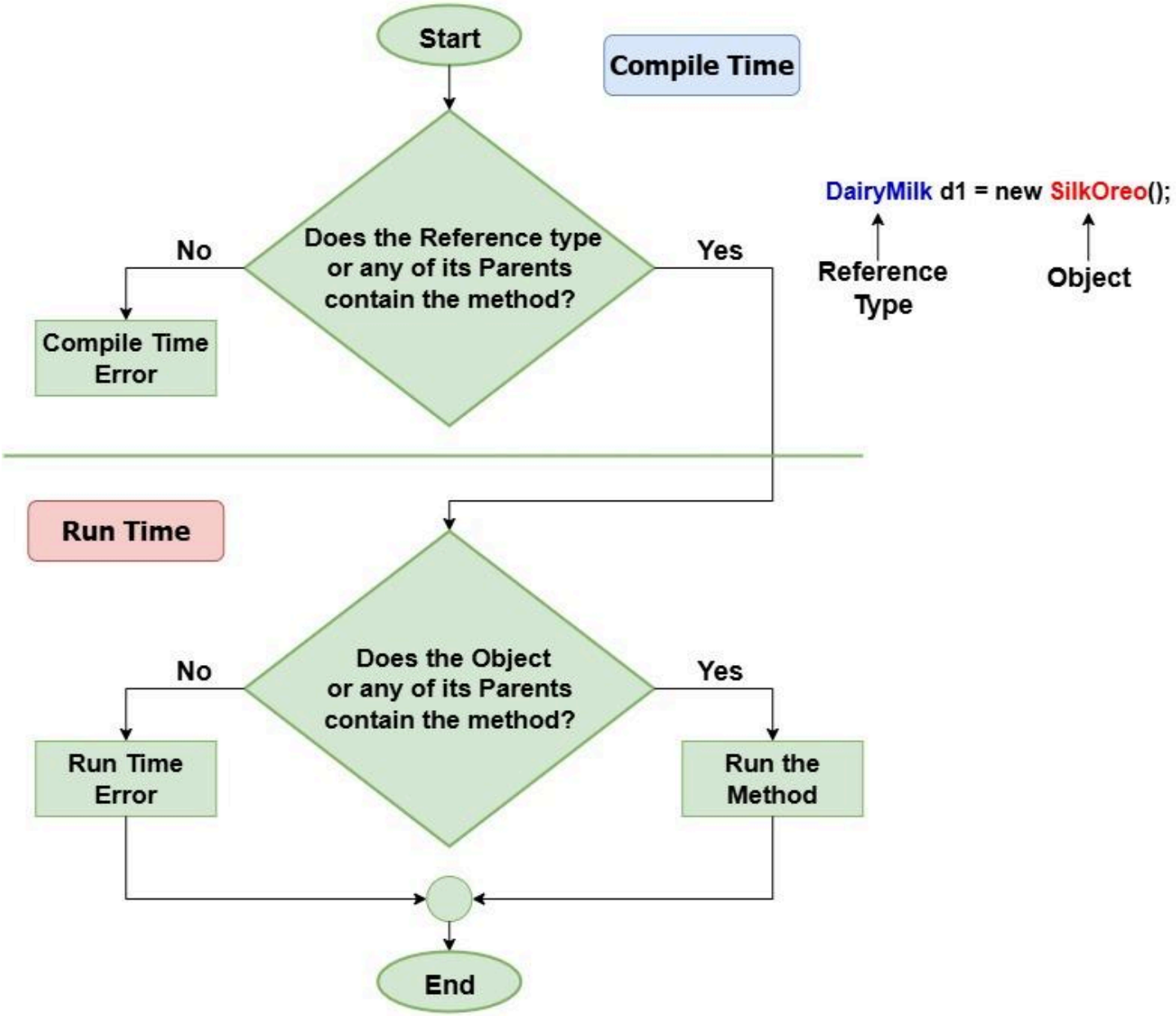
```
public class SilkOreo extends DairyMilk{
  String texture = "Silky";
  public void method1() {
    super.method1();
    System.out.println("SilkOreo m1");
  }
  public void method3() {
    System.out.println("SilkOreo m3");
    System.out.println(this);
  }
}
```

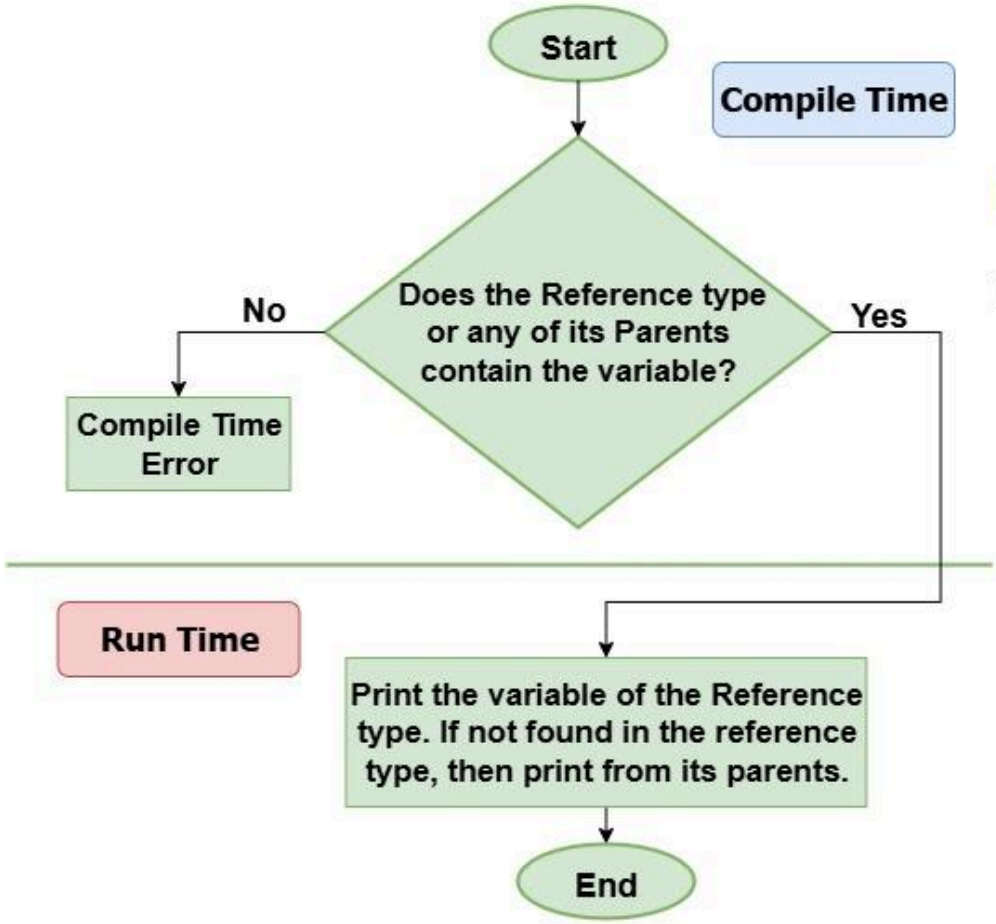
And assuming the following variables have been defined:

```
Chocolate choco1 = new Chocolate();
KitKat kit = new KitKat();
DairyMilk dairyMilk1 = new DairyMilk();
DairyMilk dairyMilk2 = new SilkOreo();
Object obj1 = new DairyMilk();
Object obj2 = new KitKat();
Chocolate caramel1 = new Caramel();
```

In the table below, indicate in the right-hand column the output produced by the statement in the left-hand column. If the statement produces more than one line of output, indicate the line breaks with slashes as in "a/b/c", which indicates three lines of output with "a" followed by "b" followed by "c". If the statement causes an error, fill in the right-hand column with either "CT" for "compile time error" or RE for "runtime error" to indicate when the error would be detected.

Statement	Output
`choco1.method1();`	
`dairyMilk1.method1();`	
`dairyMilk2.method4();`	
`caramel1.method1();`	
`System.out.println(caramel1);`	
`System.out.println(caramel1.texture);`	
`((Chocolate)kit).method2();`	
`((SilkOreo)dairyMilk2).method3();`	
`((DairyMilk)kit).method2();`	
`((Chocolate)kit).method3();`	
`((Chocolate)dairyMilk2).method1();`	
`((Chocolate)obj1).method2();`	
`((Caramel)obj1).method2();`	
`((SilkOreo)obj2).method3();`	
`System.out.println(((Object)choco1).toString());`	
`System.out.println(((Chocolate)kit).texture);`	

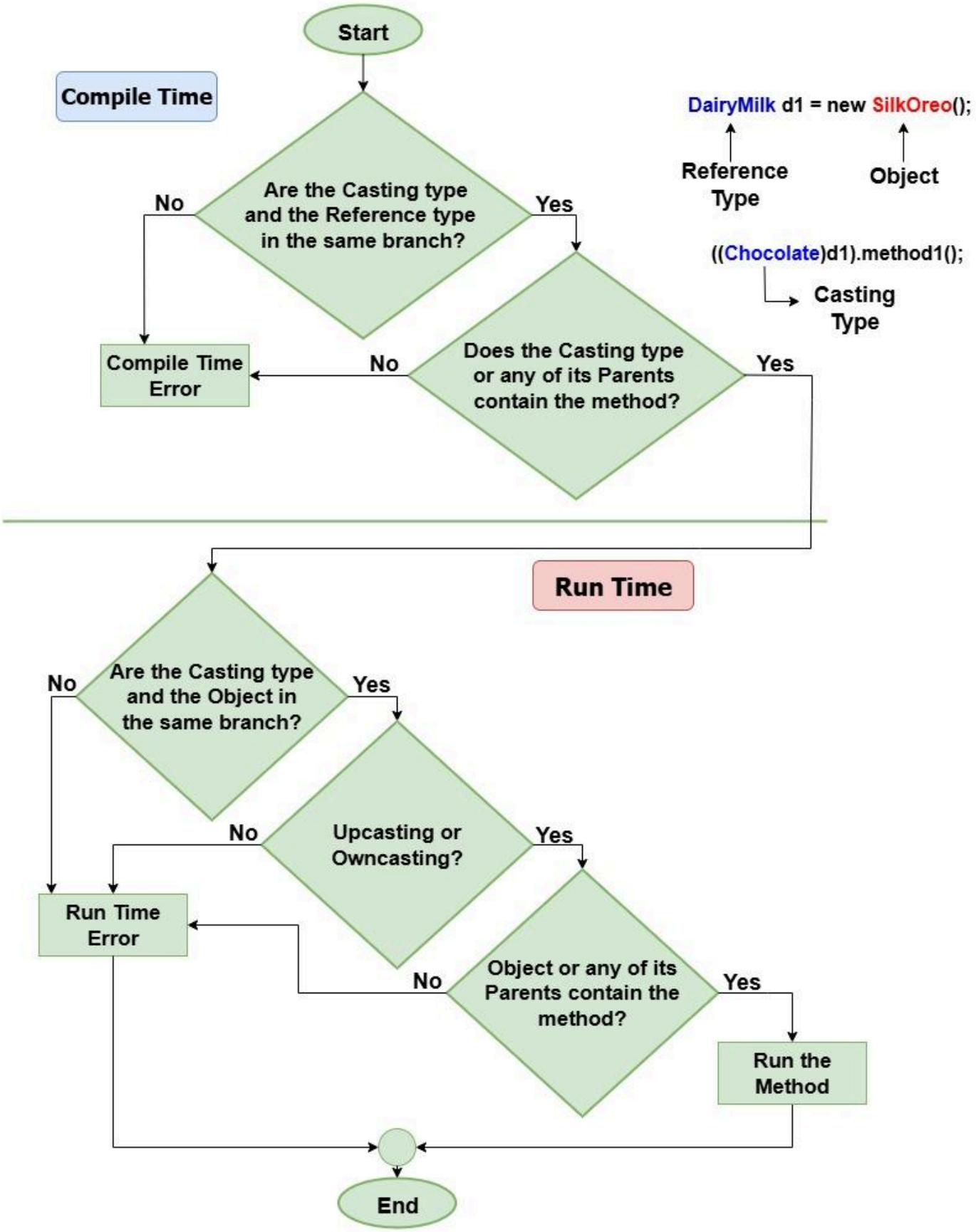

Start
Compile Time
DairyMilk d1 = new SilkOreo();
Reference Type
Object
Does the Reference type or any of its Parents contain the method?
No
Yes
Compile Time Error
Run Time
Does the Object or any of its Parents contain the method?
No
Yes
Run Time Error
Run the Method
End

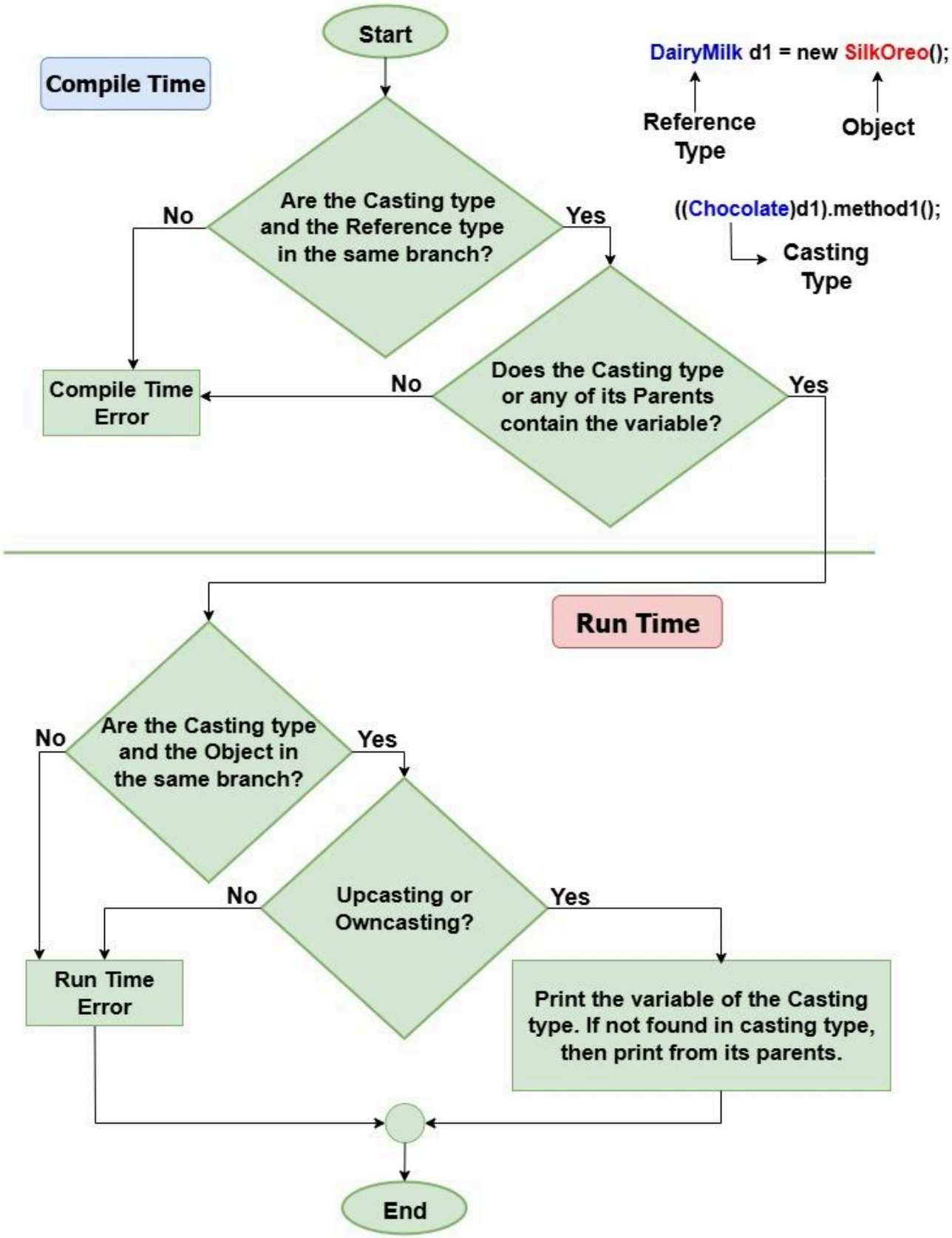



```
DairyMilk d1 = new SilkOreo();
```

Reference Type

Object


Start
Compile Time
Are the Casting type and the Reference type in the same branch?
No
Yes
Compile Time Error
Does the Casting type or any of its Parents contain the method?
No
Yes
DairyMilk d1 = new SilkOreo();
Reference Type
Object
((Chocolate)d1).method1();
Casting Type
Run Time
Are the Casting type and the Object in the same branch?
No
Yes
Upcasting or Owncasting?
No
Yes
Run Time Error
Object or any of its Parents contain the method?
No
Yes
Run the Method
End


Start
Compile Time
DairyMilk d1 = new SilkOreo();
Reference Type
Object
((Chocolate)d1).method1();
Casting Type
Are the Casting type and the Reference type in the same branch?
No
Yes
Compile Time Error
Does the Casting type or any of its Parents contain the variable?
No
Yes
Run Time
Are the Casting type and the Object in the same branch?
No
Yes
Upcasting or Owncasting?
No
Yes
Run Time Error
Print the variable of the Casting type. If not found in casting type, then print from its parents.
End